

YUVRAJ PAWAR

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PROFESSIONAL SUMMARY

Computer Science undergraduate who enjoys breaking down complex technical problems and understanding how things work under the hood. Actively expanding my expertise in full-stack development, machine learning, and DSA. Looking for an internship opportunity to contribute to hands-on engineering projects.

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, Java

Frontend: React, HTML5, CSS3, Tailwind

Backend: Node.js, Express, Flask

Databases: MongoDB, PostgreSQL, MySQL

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, Network Security

Tools: Git, GitHub, REST APIs, Postman, Render

PROJECTS

Clinic Management System

React, Node.js, Express.js, PostgreSQL, Socket.IO, JWT, Resend

- Built and deployed a full-stack clinic management platform using React, Node.js, and PostgreSQL.
- Implemented JWT authentication, role-based access control, and appointment scheduling workflows.
- Integrated Socket.IO for real-time queue updates and appointment synchronization.
- Added prescription management, email notifications, and automated appointment reminders.

EcoSimple – Environmental Monitoring Platform

React, Flask, MySQL, Leaflet, Recharts, OpenWeatherMap API

- Built a full-stack environmental monitoring platform to visualize AQI and water quality metrics.
- Integrated OpenWeatherMap API data for 10+ cities with interactive pollution heatmaps using Leaflet.
- Created dashboards using Recharts to analyze pollution trends and pollutant comparisons.

Generic Medicine Recommender

Node.js, React, MongoDB, Tesseract OCR, Groq LLM

- Developed an OCR-driven system to extract medicine names from prescription images.
- Integrated AI-based text analysis to recommend cost-effective generic alternatives.
- Designed RESTful APIs and structured MongoDB schemas for efficient query performance.

Railway Track Fault Detection Research

YOLOv8, ResNet18, PyTorch, OpenCV

- Conducted a comparative study of classification and object detection approaches for automated railway track fault detection.
- Implemented ResNet18 for fault classification and YOLOv8 for real-time defect localization using bounding boxes.
- Improved model robustness using data augmentation techniques including flipping, brightness variation, and noise injection.
- Developed a pseudo-labeling pipeline to leverage unlabeled datasets and improve cross-dataset generalization.

EDUCATION

Diploma in Computer Science

2023 – 2026

Dr. Vishwanath Karad MIT World Peace University

Data Structures · DBMS · Operating Systems · Computer Networks · Software Engineering

B.Tech in Computer Science

2026 – 2029